DBMS Lab Assignment 5 - Solution

Tables: Client_Master, Sales_Order, Products_Master. Each question shows the Oracle SQL followed by its output.

*Note: outputs were produced by running the equivalent statements on a SQLite database holding the same tables, constraints and data, and formatted like SQL*Plus. Error codes (ORA-xxxxx) are the ones Oracle raises for the same violations.*

Create the tables (all constraints named)

SQL

```
CREATE TABLE Client_Master (
  Client_no  VARCHAR2(5)  CONSTRAINT pk_client PRIMARY KEY
                          CONSTRAINT chk_client_no CHECK (Client_no LIKE 'C%'),
  Name       VARCHAR2(20) CONSTRAINT nn_client_name NOT NULL
                          CONSTRAINT uq_client_name UNIQUE,
  Address1   VARCHAR2(30),
  State      VARCHAR2(30),
  City       VARCHAR2(15) CONSTRAINT chk_client_city
                          CHECK (City IN ('Delhi','Mumbai','Chennai')));

CREATE TABLE Products_Master (
  Product_no  VARCHAR2(10) CONSTRAINT pk_product PRIMARY KEY
                           CONSTRAINT chk_product_no CHECK (Product_no LIKE 'P%'),
  Description VARCHAR2(20) CONSTRAINT nn_prod_desc NOT NULL
                           CONSTRAINT uq_prod_desc UNIQUE,
  Qty_on_hand NUMBER(8)    CONSTRAINT chk_qty CHECK (Qty_on_hand > 10),
  Sell_price  NUMBER(8,2)  CONSTRAINT nn_sell NOT NULL,
  Cost_price  NUMBER(8,2)  CONSTRAINT nn_cost NOT NULL);

CREATE TABLE Sales_Order (
  S_order_no   VARCHAR2(10) CONSTRAINT pk_sales PRIMARY KEY
                            CONSTRAINT chk_sorder_no CHECK (S_order_no LIKE 'O%'),
  S_order_date DATE,
  Client_no    VARCHAR2(5)  CONSTRAINT fk_sales_client
                            REFERENCES Client_Master(Client_no),
  Salesman_no  VARCHAR2(10) CONSTRAINT chk_salesman_no CHECK (Salesman_no LIKE 'S%'),
  Product_no   VARCHAR2(10) CONSTRAINT fk_sales_product
                            REFERENCES Products_Master(Product_no));
(Products_Master is created before Sales_Order because Sales_Order references it.)
```

OUTPUT

```
Table created.
Table created.
Table created.
```

Insert the data

SQL

```
INSERT INTO Client_Master VALUES ('C01','Ivaan','Church Rd','Maharashtra','Mumbai');
INSERT INTO Client_Master VALUES ('C02','Vandana','St.Mary Rd','Tamil Nadu','Chennai');
INSERT INTO Client_Master VALUES ('C03','Pramada','Mall Rd','Maharashtra','Mumbai');
INSERT INTO Client_Master VALUES ('C04','Basu','Church Rd','Maharashtra','Mumbai');
INSERT INTO Client_Master VALUES ('C05','Ravi','Chandni',NULL,'Delhi');
INSERT INTO Client_Master VALUES ('C06','Rukmini','Mall Rd','Maharashtra','Mumbai');
INSERT INTO Products_Master VALUES ('P01','1.44 Floppies',100,525,500);
INSERT INTO Products_Master VALUES ('P02','Monitors',25,12000,11280);
INSERT INTO Products_Master VALUES ('P03','Mouse',20,1050,1000);
INSERT INTO Products_Master VALUES ('P04','1.22 floppies',100,525,500);
INSERT INTO Products_Master VALUES ('P05','Keyboards',15,3150,3050);
INSERT INTO Products_Master VALUES ('P06','Cd drive',14,5250,5100);
INSERT INTO Sales_Order VALUES ('O19001',DATE '1996-01-12','C01','S01','P01');
INSERT INTO Sales_Order VALUES ('O19002',DATE '1996-01-25','C02','S02','P02');
INSERT INTO Sales_Order VALUES ('O19003',DATE '1996-02-18','C03','S03','P03');
INSERT INTO Sales_Order VALUES ('O19004',DATE '1996-04-03','C01','S01','P04');
INSERT INTO Sales_Order VALUES ('O19005',DATE '1996-05-20','C04','S02','P05');
INSERT INTO Sales_Order VALUES ('O19006',DATE '1996-05-24','C05','S04','P06');
COMMIT;
```

OUTPUT

```
18 rows created. (6 each in Client_Master, Products_Master, Sales_Order)

CLIENT_NO  NAME     ADDRESS1    STATE        CITY
---------  -------  ----------  -----------  -------
C01        Ivaan    Church Rd   Maharashtra  Mumbai
C02        Vandana  St.Mary Rd  Tamil Nadu   Chennai
C03        Pramada  Mall Rd     Maharashtra  Mumbai
C04        Basu     Church Rd   Maharashtra  Mumbai
C05        Ravi     Chandni                  Delhi
C06        Rukmini  Mall Rd     Maharashtra  Mumbai

6 rows selected.

PRODUCT_NO  DESCRIPTION    QTY_ON_HAND  SELL_PRICE  COST_PRICE
----------  -------------  -----------  ----------  ----------
P01         1.44 Floppies  100          525         500
P02         Monitors       25           12000       11280
P03         Mouse          20           1050        1000
P04         1.22 floppies  100          525         500
P05         Keyboards      15           3150        3050
P06         Cd drive       14           5250        5100

6 rows selected.

S_ORDER_NO  S_ORDER_DATE  CLIENT_NO  SALESMAN_NO  PRODUCT_NO
----------  ------------  ---------  -----------  ----------
O19001      12-JAN-96     C01        S01          P01
O19002      25-JAN-96     C02        S02          P02
O19003      18-FEB-96     C03        S03          P03
O19004      03-APR-96     C01        S01          P04
O19005      20-MAY-96     C04        S02          P05
O19006      24-MAY-96     C05        S04          P06

6 rows selected.
```

1. Add a NOT NULL constraint on Address1 of Client_Master and display the structure

SQL

```
ALTER TABLE Client_Master MODIFY (Address1 CONSTRAINT nn_client_address NOT NULL);
DESC Client_Master;
```

OUTPUT

```
Table altered.

SQL> DESC Client_Master
Name       Null?     Type
---------  --------  ------------
CLIENT_NO  NOT NULL  VARCHAR2(5)
NAME       NOT NULL  VARCHAR2(20)
ADDRESS1   NOT NULL  VARCHAR2(30)
STATE                VARCHAR2(30)
CITY                 VARCHAR2(15)
```

2. Profit (Sell_price - Cost_price) from Products_Master

SQL

```
SELECT Product_no, Description, Sell_price, Cost_price,
       Sell_price - Cost_price AS Profit
FROM   Products_Master;
```

OUTPUT

```
PRODUCT_NO  DESCRIPTION    SELL_PRICE  COST_PRICE  PROFIT
----------  -------------  ----------  ----------  ------
P01         1.44 Floppies  525         500         25
P02         Monitors       12000       11280       720
P03         Mouse          1050        1000        50
P04         1.22 floppies  525         500         25
P05         Keyboards      3150        3050        100
P06         Cd drive       5250        5100        150

6 rows selected.
```

3. Total cost price (Qty_on_hand * Cost_price) of stock in hand

SQL

```
SELECT Product_no, Description, Qty_on_hand, Cost_price,
       Qty_on_hand * Cost_price AS Total_Cost_Price
FROM   Products_Master;

-- grand total of the whole stock
SELECT SUM(Qty_on_hand * Cost_price) AS Total_Stock_Cost FROM Products_Master;
```

OUTPUT

```
PRODUCT_NO  DESCRIPTION    QTY_ON_HAND  COST_PRICE  TOTAL_COST_PRICE
----------  -------------  -----------  ----------  ----------------
P01         1.44 Floppies  100          500         50000
P02         Monitors       25           11280       282000
P03         Mouse          20           1000        20000
P04         1.22 floppies  100          500         50000
P05         Keyboards      15           3050        45750
P06         Cd drive       14           5100        71400

6 rows selected.

TOTAL_STOCK_COST
----------------
519150

1 row selected.
```

4. Clients whose name starts with I

SQL

```
SELECT * FROM Client_Master
WHERE  Name LIKE 'I%';
```

OUTPUT

```
CLIENT_NO  NAME   ADDRESS1   STATE        CITY
---------  -----  ---------  -----------  ------
C01        Ivaan  Church Rd  Maharashtra  Mumbai

1 row selected.
```

5. Clients whose name starts with R and ends with i

SQL

```
SELECT * FROM Client_Master
WHERE  Name LIKE 'R%i';
```

OUTPUT

```
CLIENT_NO  NAME     ADDRESS1  STATE        CITY
---------  -------  --------  -----------  ------
C05        Ravi     Chandni               Delhi
C06        Rukmini  Mall Rd   Maharashtra  Mumbai

2 rows selected.
```

6. Clients whose name has 'a' in the 3rd and 5th position

SQL

```
SELECT * FROM Client_Master
WHERE  Name LIKE '__a_a%';
```

OUTPUT

```
CLIENT_NO  NAME     ADDRESS1  STATE        CITY
---------  -------  --------  -----------  ------
C03        Pramada  Mall Rd   Maharashtra  Mumbai

1 row selected.
```

7. Clients whose name contains 'aa'

SQL

```
SELECT * FROM Client_Master
WHERE  Name LIKE '%aa%';
```

OUTPUT

```
CLIENT_NO  NAME   ADDRESS1   STATE        CITY
---------  -----  ---------  -----------  ------
C01        Ivaan  Church Rd  Maharashtra  Mumbai

1 row selected.
```

8. Clients whose name has exactly four characters

SQL

```
SELECT * FROM Client_Master
WHERE  Name LIKE '____';
```

OUTPUT

```
CLIENT_NO  NAME  ADDRESS1   STATE        CITY
---------  ----  ---------  -----------  ------
C04        Basu  Church Rd  Maharashtra  Mumbai
C05        Ravi  Chandni                 Delhi

2 rows selected.
```

9. Clients who have not mentioned a state

SQL

```
SELECT * FROM Client_Master
WHERE  State IS NULL;
```

OUTPUT

```
CLIENT_NO  NAME  ADDRESS1  STATE  CITY
---------  ----  --------  -----  -----
C05        Ravi  Chandni          Delhi

1 row selected.
```

10. Order details placed after January 1996

SQL

```
SELECT * FROM Sales_Order
WHERE  S_order_date > DATE '1996-01-31';
```

OUTPUT

```
S_ORDER_NO  S_ORDER_DATE  CLIENT_NO  SALESMAN_NO  PRODUCT_NO
----------  ------------  ---------  -----------  ----------
O19003      18-FEB-96     C03        S03          P03
O19004      03-APR-96     C01        S01          P04
O19005      20-MAY-96     C04        S02          P05
O19006      24-MAY-96     C05        S04          P06

4 rows selected.
```

Note: 'After January 1996' is read as 1 February 1996 onwards, so the two January orders (O19001, O19002) are excluded.

11. Change S_order_date to 24/07/96, Product_no to 'P06', Salesman_no to 'S04' for client 'C01'

SQL

```
UPDATE Sales_Order
SET    S_order_date = TO_DATE('24/07/96','DD/MM/RR'),
       Product_no   = 'P06',
       Salesman_no  = 'S04'
WHERE  Client_no = 'C01';

COMMIT;
SELECT * FROM Sales_Order;
```

OUTPUT

```
2 rows updated.

S_ORDER_NO  S_ORDER_DATE  CLIENT_NO  SALESMAN_NO  PRODUCT_NO
----------  ------------  ---------  -----------  ----------
O19001      24-JUL-96     C01        S04          P06
O19002      25-JAN-96     C02        S02          P02
O19003      18-FEB-96     C03        S03          P03
O19004      24-JUL-96     C01        S04          P06
O19005      20-MAY-96     C04        S02          P05
O19006      24-MAY-96     C05        S04          P06

6 rows selected.
```

Note: C01 has two orders (O19001 and O19004); both are updated because the WHERE clause is on Client_no.

12. Change the city of client 'C05' to 'Kolkata'

SQL

```
UPDATE Client_Master SET City = 'Kolkata' WHERE Client_no = 'C05';
```

OUTPUT

```
-- Attempt 1, with the original city constraint:
ERROR at line 1:
ORA-02290: check constraint (CHK_CLIENT_CITY) violated

-- The City rule only allows Delhi, Mumbai and Chennai, so widen it:
SQL> ALTER TABLE Client_Master DROP CONSTRAINT chk_client_city;
Table altered.
SQL> ALTER TABLE Client_Master ADD CONSTRAINT chk_client_city
       CHECK (City IN ('Delhi','Mumbai','Chennai','Kolkata'));
Table altered.

-- Attempt 2:
1 row updated.

CLIENT_NO  NAME     ADDRESS1    STATE        CITY
---------  -------  ----------  -----------  -------
C01        Ivaan    Church Rd   Maharashtra  Mumbai
C02        Vandana  St.Mary Rd  Tamil Nadu   Chennai
C03        Pramada  Mall Rd     Maharashtra  Mumbai
C04        Basu     Church Rd   Maharashtra  Mumbai
C05        Ravi     Chandni                  Kolkata
C06        Rukmini  Mall Rd     Maharashtra  Mumbai

6 rows selected.
```

Note: The original City constraint (Delhi/Mumbai/Chennai only) rejects 'Kolkata', so the constraint had to be extended first.

13. Change the field size of Client_no to 15 in every table that has it

SQL

```
ALTER TABLE Client_Master MODIFY (Client_no VARCHAR2(15));
ALTER TABLE Sales_Order   MODIFY (Client_no VARCHAR2(15));
DESC Client_Master;
DESC Sales_Order;
```

OUTPUT

```
Table altered.

Table altered.

SQL> DESC Client_Master
Name       Null?     Type
---------  --------  ------------
CLIENT_NO  NOT NULL  VARCHAR2(15)
NAME       NOT NULL  VARCHAR2(20)
ADDRESS1   NOT NULL  VARCHAR2(30)
STATE                VARCHAR2(30)
CITY                 VARCHAR2(15)

SQL> DESC Sales_Order
Name          Null?     Type
------------  --------  ------------
S_ORDER_NO    NOT NULL  VARCHAR2(10)
S_ORDER_DATE            DATE
CLIENT_NO               VARCHAR2(15)
SALESMAN_NO             VARCHAR2(10)
PRODUCT_NO              VARCHAR2(10)
```

14. Remove the record of Client_no 'C02' from Client_Master

SQL

```
DELETE FROM Client_Master WHERE Client_no = 'C02';
```

OUTPUT

```
-- Attempt 1:
ERROR at line 1:
ORA-02292: integrity constraint (FK_SALES_CLIENT) violated - child record found

-- C02 has an order (O19002) in Sales_Order. Delete the child row first:
SQL> DELETE FROM Sales_Order WHERE Client_no = 'C02';
1 row deleted.

SQL> DELETE FROM Client_Master WHERE Client_no = 'C02';
1 row deleted.

CLIENT_NO  NAME     ADDRESS1   STATE        CITY
---------  -------  ---------  -----------  -------
C01        Ivaan    Church Rd  Maharashtra  Mumbai
C03        Pramada  Mall Rd    Maharashtra  Mumbai
C04        Basu     Church Rd  Maharashtra  Mumbai
C05        Ravi     Chandni                 Kolkata
C06        Rukmini  Mall Rd    Maharashtra  Mumbai

5 rows selected.
```

Note: The foreign key fk_sales_client blocks deleting a client that still has orders, so the dependent order O19002 is removed first.

15. Remove records from Products_Master where Sell_price is between 1000 and 10,000

SQL

```
DELETE FROM Products_Master WHERE Sell_price BETWEEN 1000 AND 10000;
```

OUTPUT

```
-- Attempt 1:
ERROR at line 1:
ORA-02292: integrity constraint (FK_SALES_PRODUCT) violated - child record found

-- Matching products are P03 (1050), P05 (3150), P06 (5250); all are referenced by orders.
-- Delete the dependent orders first, then the products:
SQL> DELETE FROM Sales_Order WHERE Product_no IN
       (SELECT Product_no FROM Products_Master WHERE Sell_price BETWEEN 1000 AND 10000);
5 rows deleted.

SQL> DELETE FROM Products_Master WHERE Sell_price BETWEEN 1000 AND 10000;
3 rows deleted.

PRODUCT_NO  DESCRIPTION    QTY_ON_HAND  SELL_PRICE  COST_PRICE
----------  -------------  -----------  ----------  ----------
P01         1.44 Floppies  100          525         500
P02         Monitors       25           12000       11280
P04         1.22 floppies  100          525         500

3 rows selected.

S_ORDER_NO  S_ORDER_DATE  CLIENT_NO  SALESMAN_NO  PRODUCT_NO
----------  ------------  ---------  -----------  ----------

0 rows selected.
```

Note: BETWEEN is inclusive. Every remaining order pointed at one of the deleted products, so Sales_Order ends up empty. If you want to keep order history, define fk_sales_product with ON DELETE SET NULL instead.

16. Create a table of your own with a composite primary key

SQL

```
CREATE TABLE Course_Offering (
  Course_id  VARCHAR2(8),
  Semester   VARCHAR2(8),
  Instructor VARCHAR2(25) CONSTRAINT nn_instructor NOT NULL,
  Credits    NUMBER(1),
  CONSTRAINT pk_course_offering PRIMARY KEY (Course_id, Semester));

INSERT INTO Course_Offering VALUES ('CS101','2025-I','Dr. Rao',4);
INSERT INTO Course_Offering VALUES ('CS101','2025-II','Dr. Sen',4);
INSERT INTO Course_Offering VALUES ('DB201','2025-I','Dr. Mehta',3);
INSERT INTO Course_Offering VALUES ('DB201','2025-II','Dr. Mehta',3);
SELECT * FROM Course_Offering;
```

OUTPUT

```
Table created.

4 rows created.

COURSE_ID  SEMESTER  INSTRUCTOR  CREDITS
---------  --------  ----------  -------
CS101      2025-I    Dr. Rao     4
CS101      2025-II   Dr. Sen     4
DB201      2025-I    Dr. Mehta   3
DB201      2025-II   Dr. Mehta   3

4 rows selected.
```

17. Create another table whose foreign key is the composite primary key above

SQL

```
CREATE TABLE Student_Enrollment (
  Enroll_id    NUMBER        CONSTRAINT pk_enroll PRIMARY KEY,
  Student_name VARCHAR2(25)  CONSTRAINT nn_student_name NOT NULL,
  Course_id    VARCHAR2(8),
  Semester     VARCHAR2(8),
  Grade        VARCHAR2(2),
  CONSTRAINT fk_enroll_course FOREIGN KEY (Course_id, Semester)
             REFERENCES Course_Offering (Course_id, Semester));

INSERT INTO Student_Enrollment VALUES (1,'Anita','CS101','2025-I','A');
INSERT INTO Student_Enrollment VALUES (2,'Bikash','CS101','2025-II','B');
INSERT INTO Student_Enrollment VALUES (3,'Anita','DB201','2025-I','A');
INSERT INTO Student_Enrollment VALUES (4,'Chitra','DB201','2025-II','B+');
SELECT * FROM Student_Enrollment;

-- the FK is enforced: this fails because (CS101, 2026-I) does not exist
INSERT INTO Student_Enrollment VALUES (5,'Dev','CS101','2026-I','A');
```

OUTPUT

```
Table created.

4 rows created.

ENROLL_ID  STUDENT_NAME  COURSE_ID  SEMESTER  GRADE
---------  ------------  ---------  --------  -----
1          Anita         CS101      2025-I    A
2          Bikash        CS101      2025-II   B
3          Anita         DB201      2025-I    A
4          Chitra        DB201      2025-II   B+

4 rows selected.

ERROR at line 1:
ORA-02291: integrity constraint (FK_ENROLL_COURSE) violated - parent key not found
```